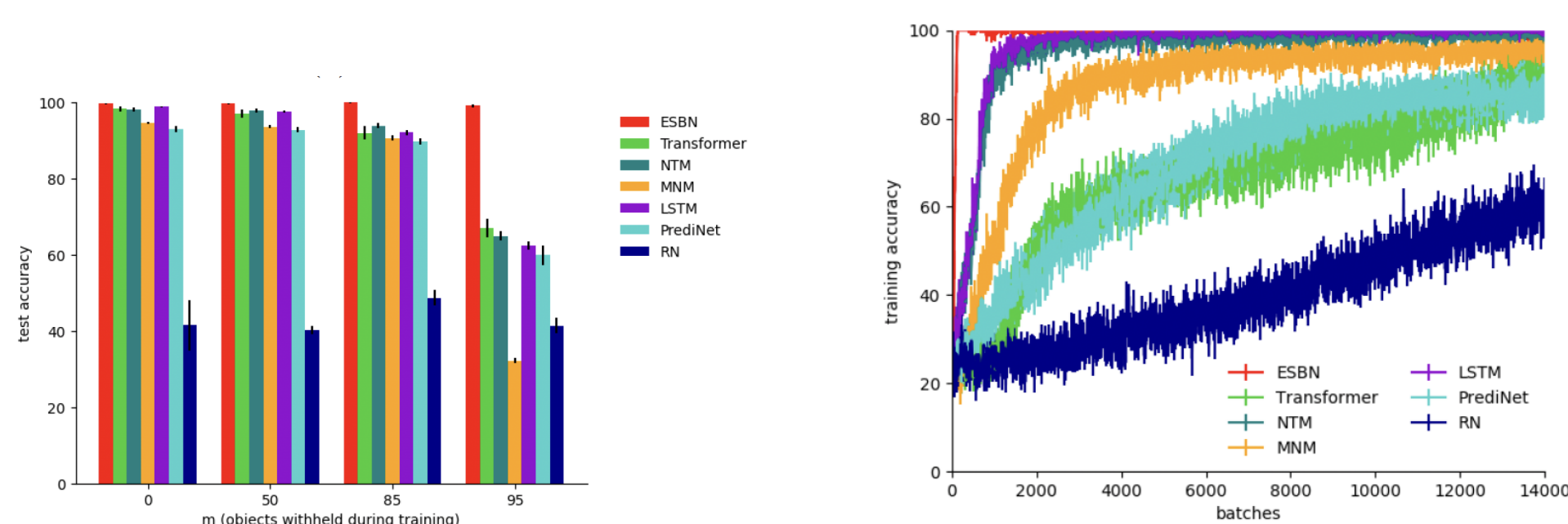


ABSTRACT

Dual Attention Transformers (DATs) augment decoder-only Transformers with a relational-attention stream that separates relational structure from sensory content. We test whether this relational inductive bias improves reasoning on ABA/ABB identity-rule tasks. Matched GPT-2 models outperform DATs despite DAT's stronger symbolic specialization. Routing analysis explains the gap: DAT can attend to the correct source without predicting it, and the relational stream contributes weakly to the output margin.

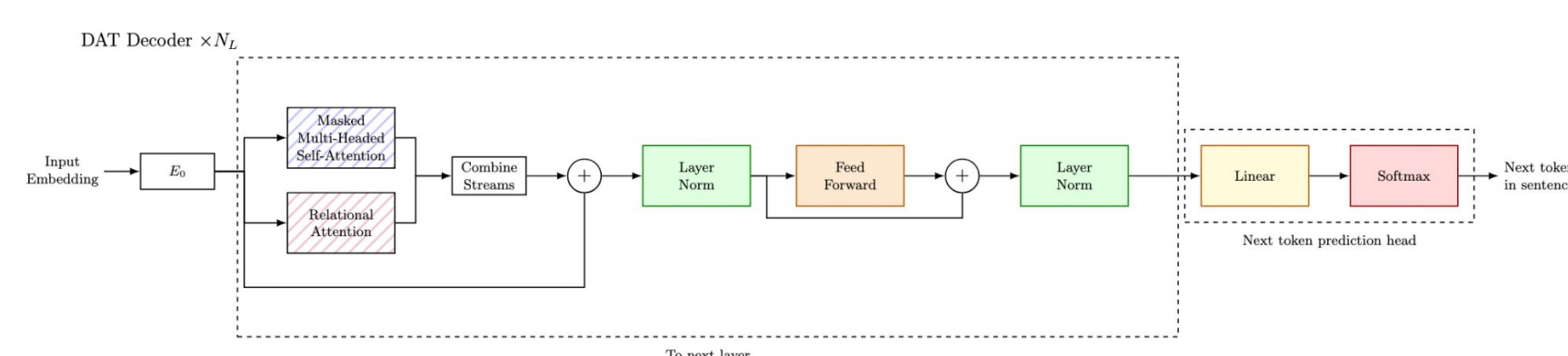
BACKGROUND

- The *relational bottleneck* prioritizes relations over irrelevant sensory detail.
- Prior relational architectures show improved OOD generalization and sample efficiency.
- This motivates testing DAT's explicit relational inductive bias against GPT-2.

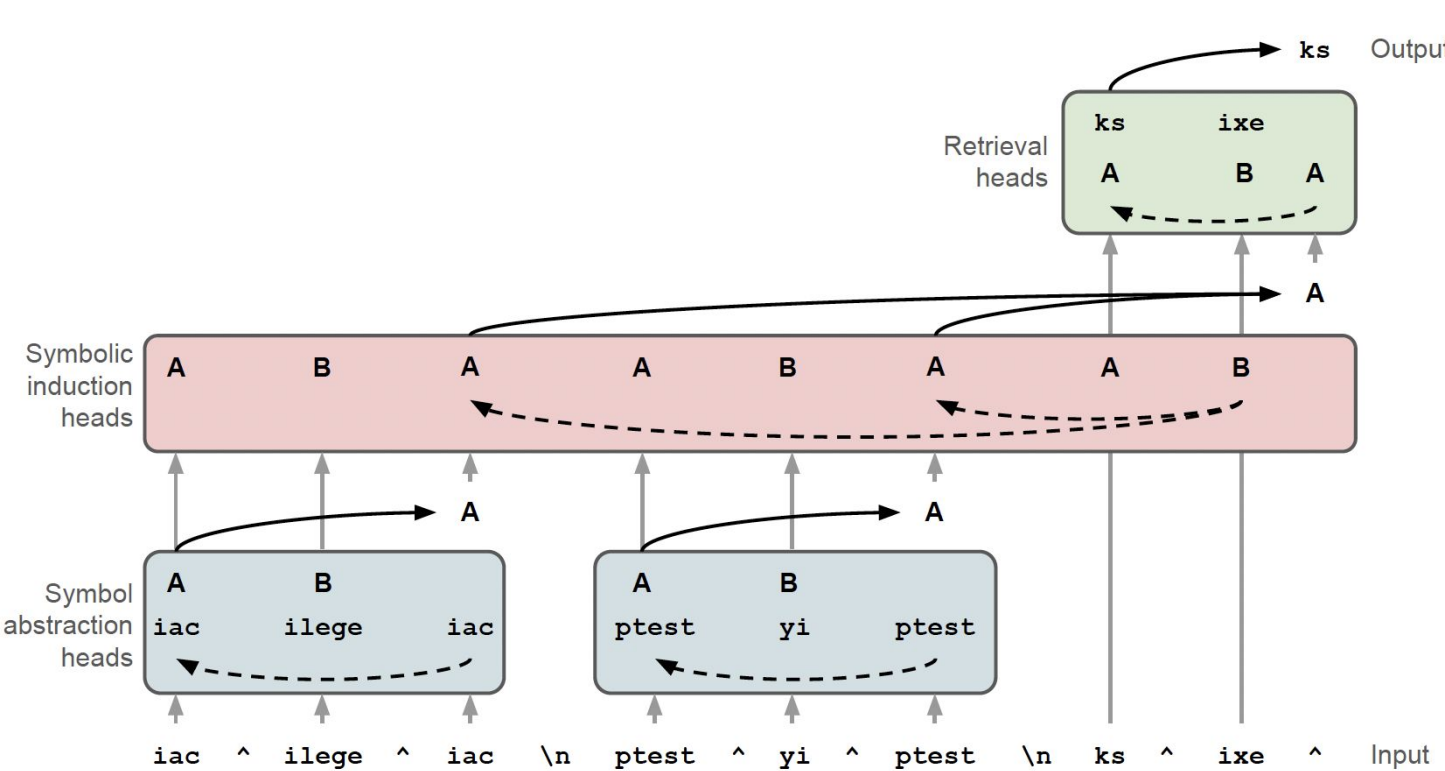


Prior ESN results: robust OOD test accuracy (left) and faster learning (right) as evidence for relational inductive bias (Webb et al., 2021).

ARCHITECTURE & MECHANISM



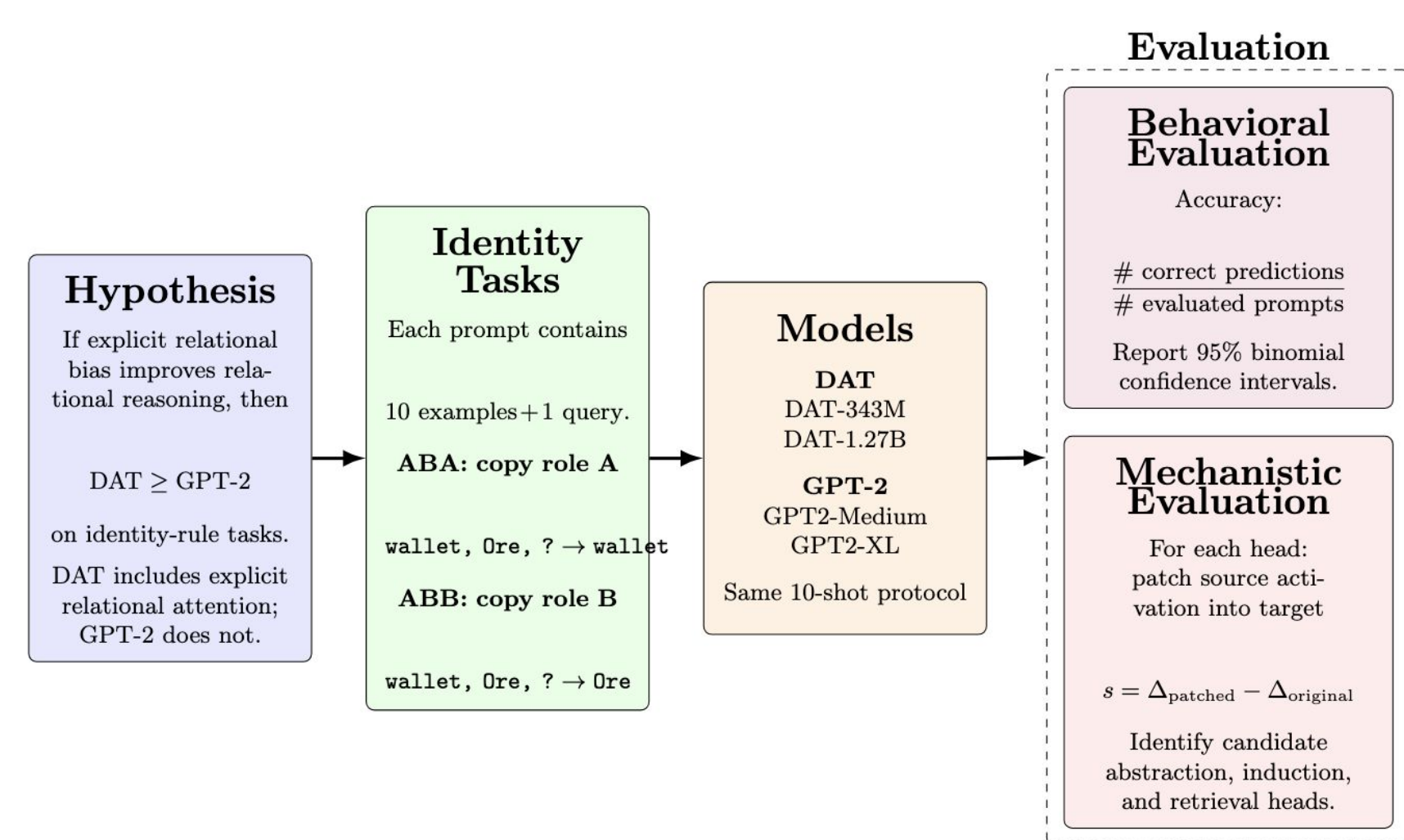
DAT decoder: self-attention + relational attention.



Symbolic circuit: abstraction $\rightarrow$ induction $\rightarrow$ retrieval.

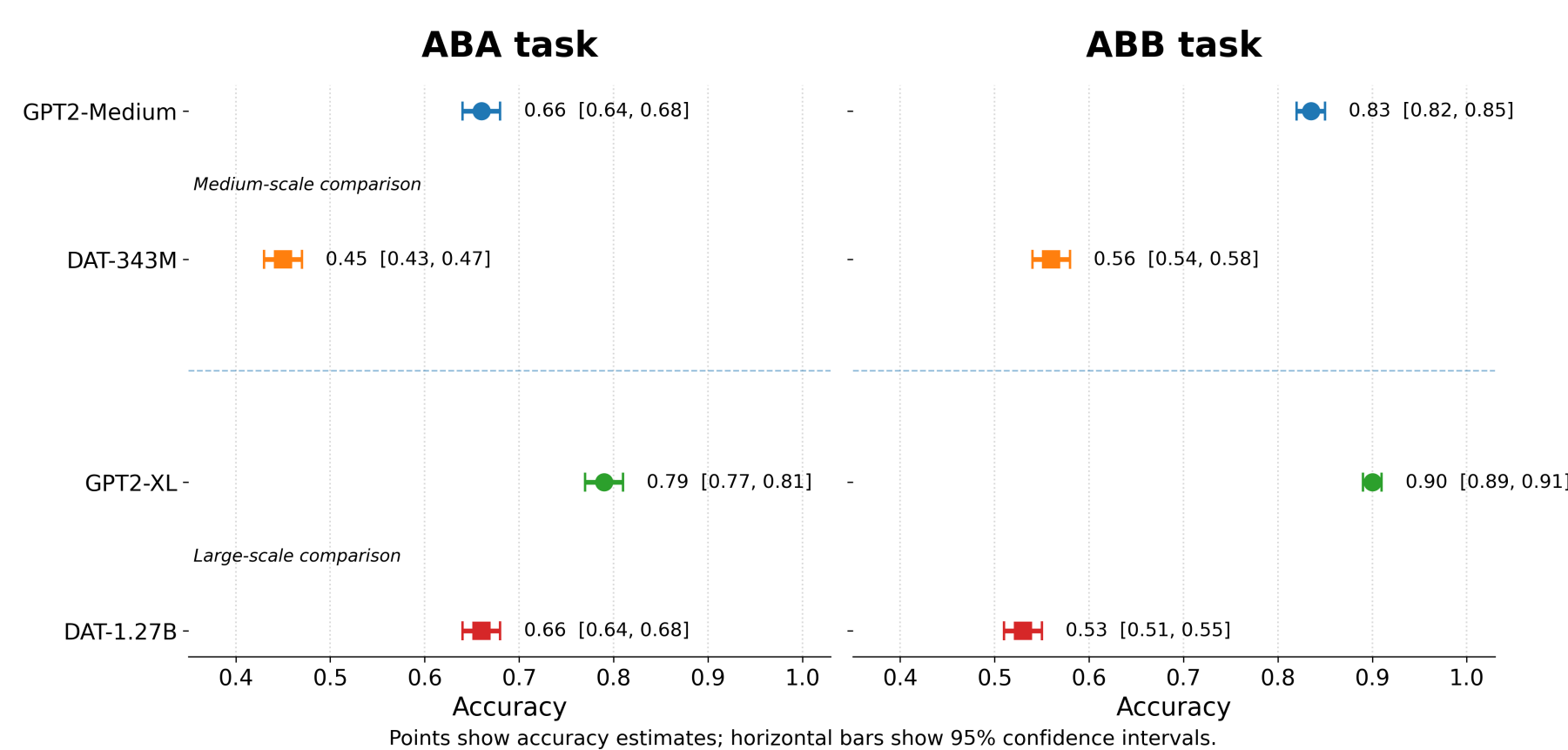
EXPERIMENTAL DESIGN

- Models:** DAT-343M/1.27B, GPT-2 Medium/XL
- Tasks:** 10-shot ABA & ABB completion
- Eval:** accuracy; CMA; source selection; routing margin; ablation/amplification



Study plan: behavior + mechanism $\rightarrow$ diagnose the gap.

GPT-2 OUTPERFORMS DAT

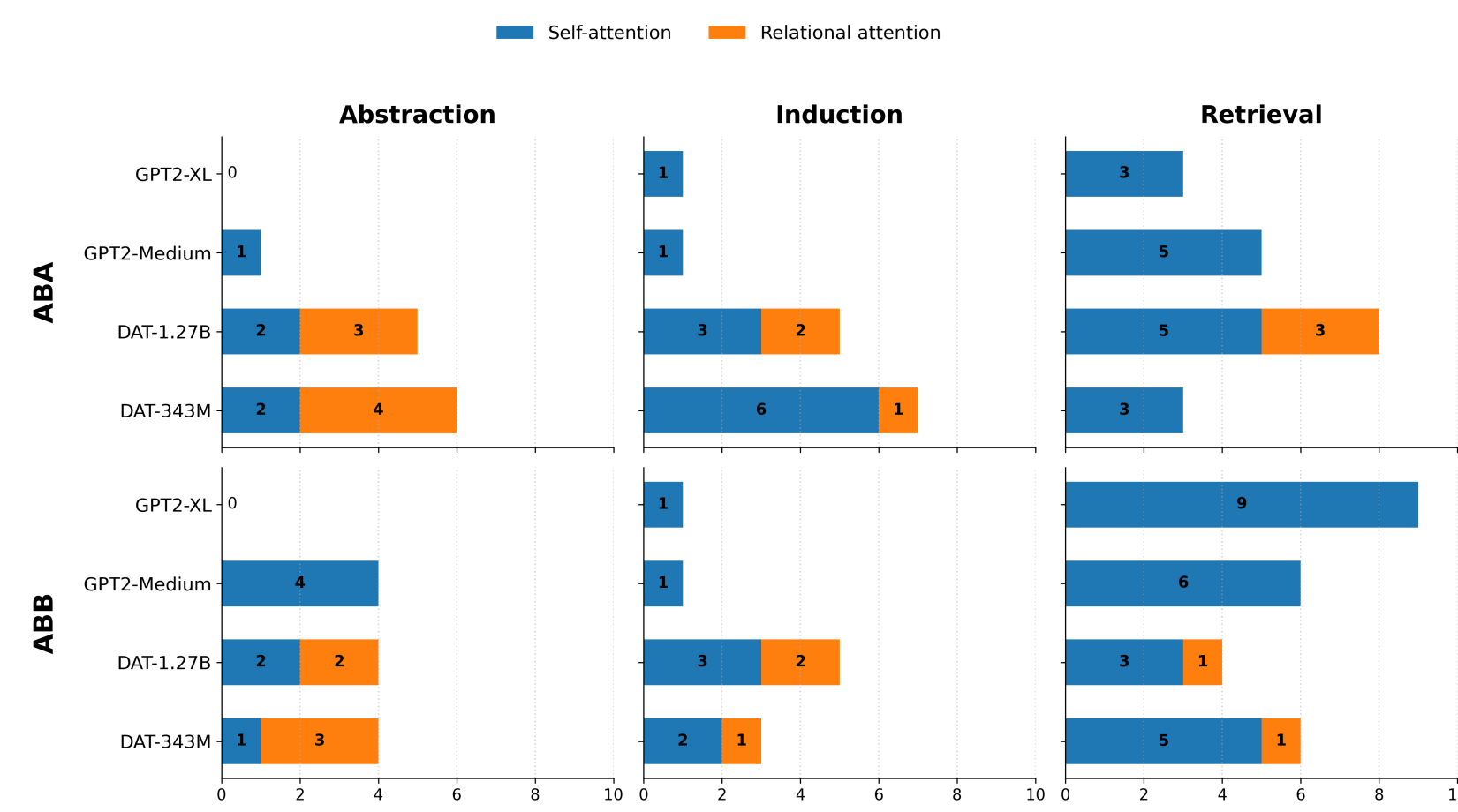


10-shot accuracy (95% CI). GPT-2 wins at both scales.

Takeaway: Explicit relational attention $\neq$ best accuracy.

SPECIALIZATION $\neq$ ACCURACY

Candidate Symbolic Heads Identified by Causal Mediation Analysis

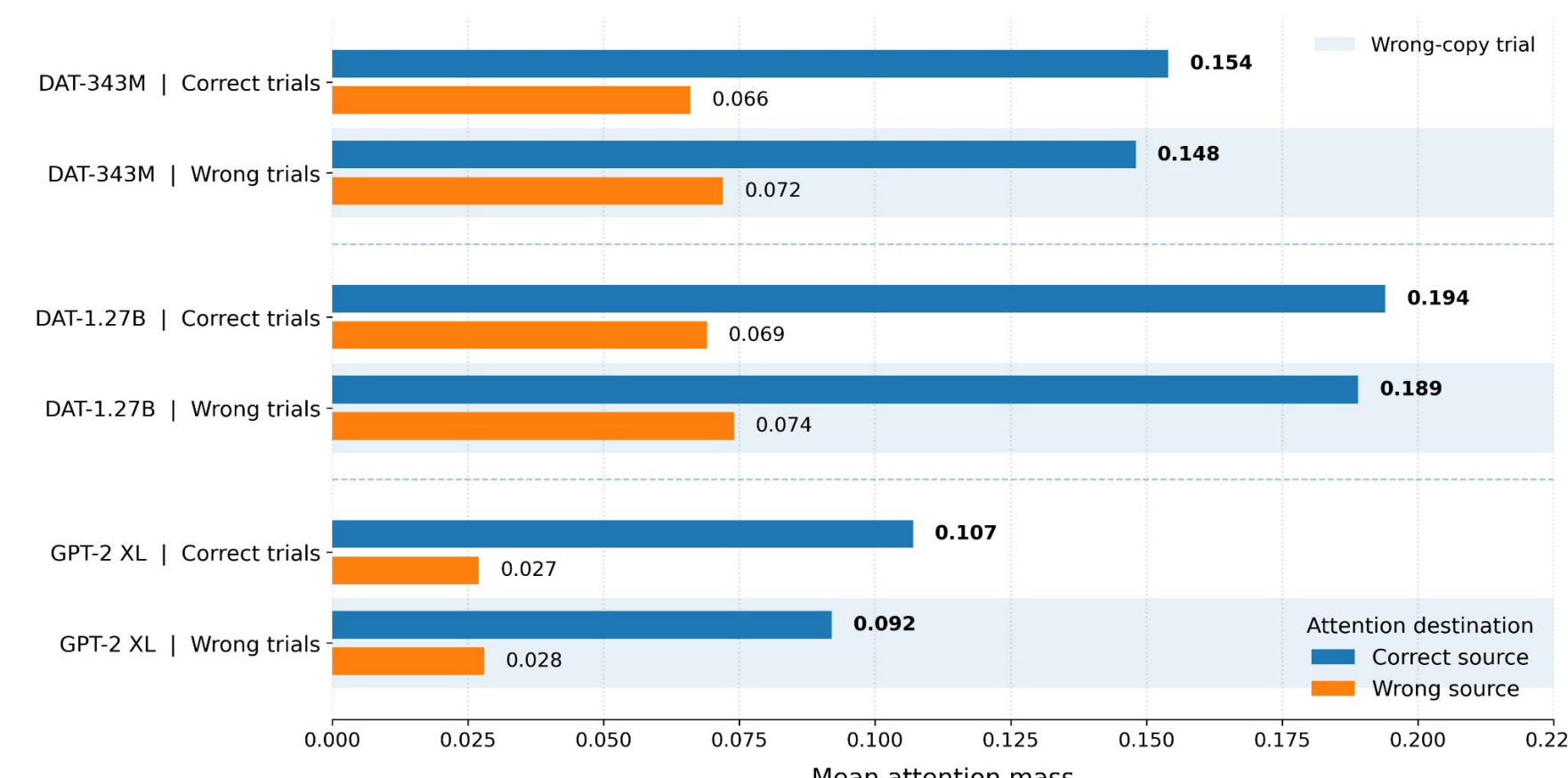


DAT exhibits broader abstraction and induction specialization, including candidate heads in the relational-attention stream.

Takeaway: DAT has broader abstraction/induction specialization.

WHY THE GAP: ROUTING

DAT often copies the *wrong* query token. Attending the right source is not enough:



On ABB wrong-copy trials, DAT relational attention still assigns more mass to the correct source than the wrong source.

Source selection (ABB): wrong-copy trials can still prefer the correct source.

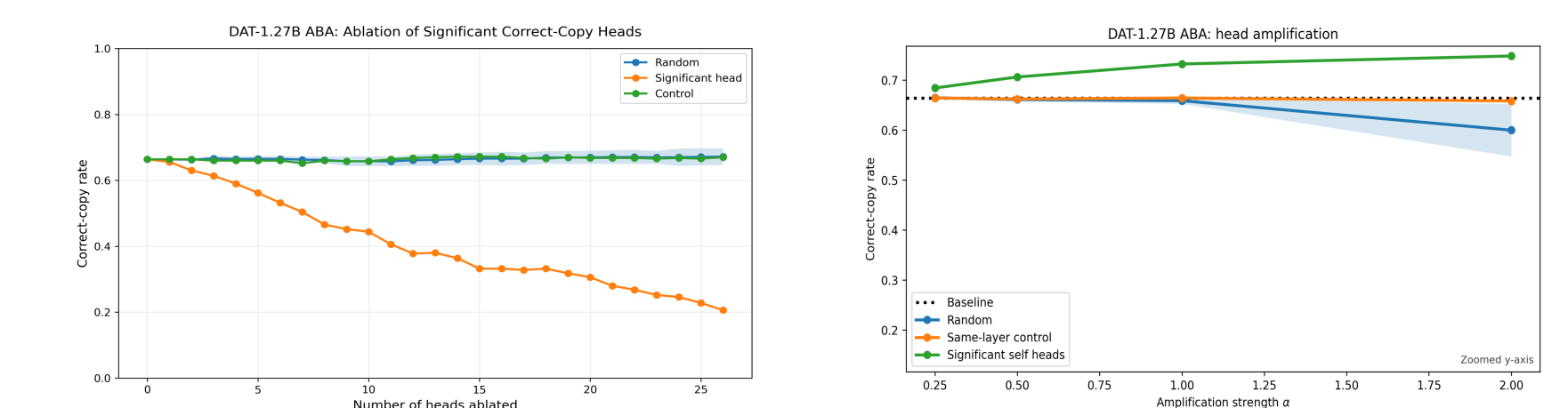
Summed positive routing contribution (correct-copy trials):

Model	Stream	ABA	ABB
GPT-2 Medium	Self	103.6	122.1
GPT-2 XL	Self	50.8	92.4
DAT-343M	Self / Rel	6.0 / 0.3	6.2 / 0.4
DAT-1.27B	Self / Rel	23.5 / 1.8	18.2 / 2.5

DAT's relational stream contributes weakly to the correct-vs-wrong margin.

Takeaway: Attention location $\neq$ useful output contribution.

ABLATION & AMPLIFICATION



DAT-1.27B ABA: ablate routing heads $\Rightarrow$ drop; amplify $\Rightarrow$ gain.

CONCLUSION & REFERENCES

- Mechanism $\neq$ behavior:** DAT specializes more; GPT-2 performs better.
- Cause:** weak routing of relational information to the output.
- Implication:** inductive bias must also shape the final prediction.

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[3] T. W. Webb et al. "The Relational Bottleneck as an Inductive Bias for Efficient Abstraction." *Trends Cogn. Sci.*, 28(9):829–843, 2024.

[4] Y. Yang et al. "Emergent Symbolic Mechanisms Support Abstract Reasoning in Large Language Models." arXiv:2502.20332, 2025.

[5] N. Elhage et al. "A Mathematical Framework for Transformer Circuits." *Transformer Circuits Thread*, 2021.